

Voice & Image Product Search + Social Ad Metrics — Design & Compliance Brief

What's new (2026-09-19). Three shipped capabilities on Get Goods Gratis (Free): (1) **voice** product search, (2) **image/photo** product search, and (3) the full advertising-metric set for the **social channel**, tracked for **every advertiser and the platform's own business ads**. All three work on the website and, through the Capacitor wrapper, on **iOS and Android**. This brief describes the mechanism and flags the (light) legal touchpoints. Not legal advice.

1. Voice product search

What it does. A shopper taps the mic in the store search bar and speaks a product name instead of typing it.

How it works. On-device first: browsers with the **Web Speech API** (`SpeechRecognition`) transcribe locally — the audio never leaves the device. Browsers without it (notably the iOS app `WebView`) fall back to recording a short clip and sending it to the server function `voiceSearchTranscribe`, which transcribes it (Whisper via the existing `transcription.ts`) and returns text. Either way the transcript drives the normal product search. The mic is used only after the user grants the browser/OS permission, and only while actively searching.

Flag: `VOICE_SEARCH_ENABLED` (on by default). **Engine:** `visual-voice-search`. **Function:** `voiceSearchTranscribe`.

2. Image / photo product search

What it does. A shopper uploads a photo (or takes one on mobile) and the platform finds that product.

How it works. The image is uploaded (existing `Core.UploadFile`), then a **vision model identifies** the product (brand + model + key attributes) via `imageProductSearch`; the identified query drives a product-feed search and returns results in the **same shape** as text search, each tagged with its sanctioned checkout channel. Image capture is entirely user-initiated (a file picker / camera).

Flag: `IMAGE_SEARCH_ENABLED` (on by default). **Engine:** `visual-voice-search`. **Function:** `imageProductSearch`.

3. Social ad metrics — all advertisers + own company

What it does. The full network-standard advertising metric set — reach, impressions, clicks, CTR, conversions, CVR, engagement, spend, revenue, **eCPM, CPM, CPP, windowed D1-D365 ROAS**, and delivered ad value — computed for the **social channel** (member-amplified posts + the platform's own AI social ads), for **every advertiser** and for the **platform's own business ads**.

How it works. The new `social-ad-metrics` engine reads the existing `SocialMediaPost` activity and computes the metric set per advertiser, as a platform-wide aggregate + a per-advertiser leaderboard, and for the platform's own ads (`post_type = "platform_own_ad"`). The `socialAdMetrics` function serves it: an advertiser sees their own; an admin can query any advertiser, the platform's own ads, the total, or the leaderboard. Surfaced on the advertiser (`AdBusinessDashboard`) and admin (`AdminDashboard`) dashboards via `SocialAdMetricsPanel`.

Honesty posture (unchanged). Every figure is **MEASURED** from real activity or clearly labeled with its basis; below the data threshold ratios read "still gathering data," never a faked number; spend-based costs (CPM/PP/ROAS) appear only when a real social spend is recorded; impressions are estimated from reach only when a post has no measured impression count (at the `SOCIAL_VIEW_RATE` estimate, labeled). **Delivered "ad value" = impressions × the conventional CPM — never a guaranteed revenue/ROI result.**

Flags: `SOCIAL_AD_METRICS_ENABLED`, `SOCIAL_AD_METRICS_MIN_POSTS`, `SOCIAL_VIEW_RATE` (all on/measured by default).

4. Legal / compliance touchpoints for counsel

- **Microphone consent (voice).** Capture uses the browser/OS permission prompt; on-device transcription keeps audio local. The server fallback sends a short clip to a transcription provider — cover it in the privacy policy / sub-processor list (Whisper/OpenAI or self-hosted). This is speech-to-text for a search box, **not** voiceprint / biometric identification, so it is outside BIPA-style biometric-identifier scope — confirm and keep it that way (no speaker recognition). See `DATA-PROCESSING-AND-SUBPROCESSOR-BRIEF`, `BIOMETRIC-CONSENT-AND-DATA-POLICY-BRIEF`.

- **Camera / photo consent (image).** User-initiated upload/camera; the image is sent to a vision provider to identify the product — disclose the provider and the purpose; don't retain the image beyond the search unless disclosed.
- **App-store permission strings.** iOS `NSMicrophoneUsageDescription / NSCameraUsageDescription / NSPhotoLibraryUsageDescription`; Android `RECORD_AUDIO / CAMERA`. See `MOBILE-APP-WRAPPER-GUIDE.md`.
- **Advertising metrics.** The measured-not-guaranteed framing already reviewed for `ad-metrics.ts` applies unchanged to the social metric set — confirm the social eCPM/CPM/ROAS wording carries no implied ROI guarantee. See `FOR-YOUR-ATTORNEY §10-11, AD-MEDIA-AND-TARGETING-DESIGN`.

Not legal advice. Reuses existing entities (no new data categories). Feature flags default on; the capture of mic/ camera is always user-permissioned.